

Machine Learning

as a Tool for Analysis in Social Sciences[†]

by

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Abstract

This paper shows how Machine Learning techniques can be used for analysis in the Social Sciences. The general paradigm and the relevance of Machine Learning is described and then a specific problem is addressed: that of classifying countries with high strife. A program is used to produce classification rules to attack that problem and the performance of the rules are tested. We show that Machine Learning techniques such as the one described here can indeed be very useful for analysis of this sort. The rules that are generated are more readable than numerical fits to data and are usually fairly concise. Moreover the reliability of the results can be very high. Finally, there is a good possibility that the methodology can be extended in the near future to effectively utilize high level domain knowledge. It is emphasized that the procedure helps to automate the *search* for a good theory and that many plausible theories are generated in the process. We envision systems of the future working in tandem with scientists, helping to generate novel theories for explaining social phenomena.

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1. Introduction

The objective of this paper is to show that techniques developed in AI and Machine Learning (ML) research (the study of machines which learn) can be useful tools in the quantitative study of social systems. Machine Learning programs have previously been used effectively to discover classification rules in several different domains (for examples, see [Buchanan, *et al.*, 1976], [Michalski and Chilausky, 1980], [Langley, *et al.*, 1983], and [Lenat, 1976].) In this study we will use a program developed at the University of Illinois AI Laboratory to study those countries which reported a high incidence of riots and armed attacks during the year 1977. Each country will be described by a set of measurements and variables and classified according to the number of incidences per million people. The idea is to try to generalize these individual event descriptions into meaningful class descriptions which can then be used to explain the data.

2. Machine Learning

A large part of Machine Learning research has been concerned with the problem of learning rules to classify events into a number of different classes, rendering it relevant to this sort of study. The general paradigm is as follows. A set of mutually exclusive classes is identified together with a set of variables presumed to be sufficient for distinguishing between the different classes. The algorithm takes a set of events described by these variables, together with the class that each event belongs to, and produces general rules in terms of these variables for classifying new events. It is hoped that the rules, or hypotheses, produced by the algorithm are good classifiers of future events.

In general there are numerous hypotheses which are consistent with a given set of examples, although there is probably only one that is going to be correct for all future examples. Since, *a priori*, the entire set of examples and their associated classes is unknown, there is no reason for choosing one consistent rule over another, hence the identification of the correct rule becomes a probabilistic event dependent on the the number of consistent

hypotheses. This, together with the fact that the number of consistent hypotheses grows exponentially with the number of variables, has the unfortunate consequence that the probability of hitting upon the right hypothesis approaches zero very quickly.

There is one way to overcome this problem. The above argument rests on the fact that there is no inherent structure to the events, and that the hypotheses can be arbitrarily complex. Usually this is not the case. The user normally has some information about the events and the probable hypotheses. This information can be used as criteria for choosing or rejecting candidate hypotheses, i.e., it can be used as a bias in the algorithm. For example, one bias which seems to work well in most domains is to identify some sort of regularities in the example set and choose hypotheses which are consistent with these regularities. Another common bias is to choose hypotheses which tend to classify events which are "similar" to each other into the same class. Here similarity between two events is defined as the extent to which the values of their variables are similar.

In addition to the above, in the program that was used in this study, there were several other ways that one could affect the rules which were chosen. It was possible to specify certain lexical criteria (such as choosing the simplest or shortest rule) which were used to order the hypotheses. One could enter domain knowledge directly in the form of logical rules which relate different variables together. One could also associate a cost function with each variable, with the rules having the minimum total cost being selected.

3. Astra

The program that we will use is called **Astra**. It is a user-friendly version of the **Induce** algorithm that was developed at the AI Laboratory at the University of Illinois at Urbana-Champaign under the direction of Prof. R.S. Michalski [Hoff, Michalski and Stepp, 1983]. Astra is a system with facilities for entering events, changing them, and generating various types of classification rules in one integrated package which greatly enhances its ease of use.

Astra takes as its input a series of events, described by a set of variables, each of which is associated with a particular class. Using these examples and the user-defined preference criteria for selecting rules, it generates a candidate list of possible hypotheses and selects the "best" one. Each rule that is generated is guaranteed to be consistent and complete with respect to the input events. Completeness ensures that all the input events will be classified under at least one of the rules, whereas consistency ensures that the rules will classify the original events into their original classes.

3.1. Discriminant and Characteristic Rules

There are two types of rules that can be generated by Astra: discriminant rules and characteristic rules. Discriminant rules classify the input events in such a way that no input event of one class is classified by a rule for another class. Characteristic rules are rules generated by generalizing the input events of a particular class without considering whether they also classify events of other classes. Both types of rules can be very useful when studying a domain. Discriminant rules are well suited for classification tasks and for predicting which class a new event would fall into. Characteristic rules are useful for identifying which variables are important to a particular class and may be used to help develop models of the domain. Both types of rules have the potential to provide novel insights into the particular problem being studied.

3.2. The Input to Astra

Events are entered into Astra as a series of *descriptors*, with each descriptor denoting one event. A descriptor has a left hand side containing a sequence of *selectors*, which denote the values of the different variables, and a right hand side which contains a selector describing the class that the event belongs to. A selector is of the form [term = value(s)] and corresponds roughly to a predicate in predicate calculus. The term can be either a variable or a function of several variables (denoting a relationship between the variables). For example, the following are legal selectors:

[literacy(c1)=19,20]	- c1 has a literacy value of 19 or 20
[next_to(c1,c2)=true]	- c1 and c2 are characterized by the relationship next_to.

The following is an example event descriptor:

[literacy(c1)=19,20][sanctions(c1)=0] => [strife = low].

This represents the fact that a country with a literacy value of 19 or 20, and whose government has imposed no sanctions is classified as having low strife. Note that *all* the selectors on the left hand side have to be true in order for the rule to be satisfied, i.e., the selectors represent a conjunction of predicates.

The Induce algorithm, which is used to generalize rules of the above form, and its various properties are described in detail in [Hoff, Michalski, and Stepp, 1983].

3.3. Previous work with Induce

Induce has previously been used to classify soy bean diseases [Michalski and Chilausky 1980]. A number of examples were supplied and rules for diagnosing diseased soy beans were generated from them. The performance of these rules were then tested against a set of rules provided by a human expert. It is worthwhile to note that the rules generated by Induce outperformed the rules provided by the expert. Thus the Induce algorithm has already shown itself to be an effective classification tool in at least one domain.

4. The Relevance of Astra to Analysis in Political Science

There are a number of reasons why Astra is particularly well suited to this study:

- The generated rules are complete, i.e., every input event is classified under one of the rules.
- The generated rules are consistent, i.e., the rules that are generated are consistent with the example events. The rules classify the original events into their original classes.
- Induce can automatically determine which variables are relevant to the particular set of examples given. This can be influenced by the user by associating costs with each variable. The variables with lower cost are considered to be more important; the algorithm attempts to include these variables in the final rule.
- The program can handle a large number of variables at one time (limited only by the resources available). This is a strong advantage in domains, such as the one being considered here, where there are a large number of potentially important variables and it is not known a priori which ones are relevant.
- The rules that are generated are easy to read, easy to test on further data, and meaningful to humans. This is not true of many other ML programs and constitutes a major advantage. If the rules are to be used later for tasks which have important consequences, it is essential to be able to check these rules by hand. Moreover, as in this particular case, the readability of the rules allows one to examine them easily and develop new theories about the domain.

- Lastly, the algorithm is not just a formula that you plug your data into. One can affect the rules that are generated by entering higher level knowledge about the domain and how the different variables relate to one another.¹

5. Coding of the data

The source of data for this work is from [Taylor and Jodice, 1983], a two-volume set which contains the data for numerous important political indicators. All of the variables that were used are either taken directly from the book or derived from some of the indicators in the book. An attempt was made to choose a wide variety of different variables representing different types of indicators. In particular, there were variables representing the extent of governmental restraints (polright and civright), the population density (popul), presence of social mobilization (highered and literacy), the economic structure (mml, mgdp, kind), state coercive behavior (sanctions, relaxsanc), social discontent (polstrikes), and geographical location (continent). Also included were variables which measured changes in some of the above values (prshift, crshift, enshift, and litshift). Each variable is described in more detail below.

Altogether there were fifty-one countries in the data set each classified as being either high or low strife countries during the year 1977 (see definition below). Twenty-eight of the countries were classified as low strife countries and twenty-three were classified as high strife.

5.1. The variables

The following variables were used to describe the countries:

¹Although it is possible to do this in Induce to a limited extent, the current state of research in ML is such that one is still restricted in the types of knowledge that can be conveniently entered. Expanding this capability is an important area of research in this field.

strife - This was the main variable, used to classify the countries into two sets - low and high, and was determined by the number of riots and armed attacks that were reported, per million people, during the year 1977. A riot is defined as a violent demonstration and an armed attack is defined as an act of violent political conflict carried out by an organized group. If the strife was less than 0.3, then the country was said to have low strife. If it was greater than or equal to 0.3, then it was said to have high strife.

literacy - This variable indicated the percentage of adults in a country considered to be literate. The value of the variable was the percentage divided by 5 plus 1, and ranged from 1 to 20.

popul - This variable measured the concentration of the population. It was defined as 10 times the log of the population concentration. The logarithm of the concentration was taken since the graph of this variable (with the countries ordered by concentration on the x-axis) tends to be exponential. The possible values for this variable ranged from 1 to 30.

highered - Measures the number of students enrolled in higher education, per million population. The value of the variable was :

- 0 if the number was less than 1500
- 1 if the number was between 1500 and 4099
- 2 if the number was between 4100 and 8999
- 3 if the number was greater than 9000

polright - Measured the extent of political rights in the specified country. The value of the variable is the mean value of the Political Rights Index as compiled by Raymond Gastil, [Gastil 1973-1979] during the years 1973 to 1979, rounded to the nearest integer. It varied from 1 (indicating the most rights) to 7 (indicating the least rights).

civright - Measures the extent of civil liberties in a country.

The value of this variable is the mean score of the Civil Rights Index of the specified country during the years 1973-1979, also compiled by Raymond Gastil, rounded to the nearest integer. As above, the value ranged from 1 to 7.

sanctions - Measured the number of sanctions that were imposed by the government of the specified country during 1977. A sanction is defined an action taken by the authorities to control a perceived threat to the political system. The values of the variable were:

0 if the no. of sanctions were between 0 and 5.

1 if the no. of sanctions were between 6 and 15.

2 if the no. of sanctions were between 16 and 69.

3 if the no. of sanctions were greater than 69.

relaxsanc - Measured the number of sanctions that were relaxed by the authorities. The values of the variable were the same as above.

prshift - Measured the maximum shift in the Political Rights Index during the years 1973 to 1979. The value of the variable was the shift + 6. The possible values ranged from 0 to 11.²

crshift - Measured the maximum shift in the Civil Rights Index during the years 1973 to 1979. The value of this variable was the shift + 4. The possible values ranged from 0 to 8.

enshift - Measured the shift in school enrollment for a particular country. The value of this variable is the shift plus 4. The possible values ranged from 0 to 23.

²Since negative shifts were present, and the present version of Astra is limited to positive values for the variables, the encoding for the variables indicating shift were adjusted to a positive scale. There was a further limitation in that the version of Astra that was used could only hand a maximum of 30 different values per variable.

litshift - Measured the shift in the literacy rate of a country. The value of this variable is the shift plus 18 divided by 3. The possible values ranged from 0 to 22.

continent - the value of this variable was one of the following: europe, namerica, samerica, africa, and asia. It represented the continent that a particular country was in.

polstrikes - Measured the number of political strikes that took place in the year 1977. A political strike is defined as a strike or work stoppage by a group of workers or students in protest of the government. The value of this variable was the actual number of strikes.

kind - This variable represented three economic sectors: services, industry, and agriculture. The value was either services, indus or agr.

mgdp - This variable was a predicate of two arguments (representing economic sectors) and was true if a larger share of the Gross Domestic Product of a particular country was due to the first sector.

mml - This variable was also a predicate of two arguments representing economic sectors and was true if a larger share of the manual labor in a country worked for in the first sector.

6. Results

Discriminant rules were generated by randomly selecting half of the low strife countries (14) and half of the high strife countries (12) and generating rules differentiating the two classes. The generated rules were then tested on the remaining events to see how well they performed on new events. The set of countries which were used to generate the rules will be called the sample, and the rest of the countries will be said to comprise the test set. Since the

countries were picked randomly, rules were generated with several sample sets. This tends to negate the effects of any specific sample being particularly good or bad. The results are as follows:

For both the low and high strife countries the rules performed very well on the test set. The percentage of the low strife countries that were correctly classified ranged from 79% to 93% with a mean of 86%. The percentage of the high strife countries that were correctly classified was slightly lower and ranged from 64% to 91% with a mean of 80%. We believe that the rules performed better on the low strife countries because there were probably more regularities in their descriptions than in the descriptions of the high strife countries. The rules that were produced support this argument (see below). In general however, the rules tended to be very good predictors for events that had not been seen while generating the rules.

In all cases, due to the consistency and completeness criteria, the rules were 100% correct on the sample set. Thus the average performance of the rules on the entire set of events was 93% for countries with low strife and 91% for countries with high strife.

The following is an example of a rule that was generated for countries with low strife. It correctly classified 26 out of the 28 low strife countries. Both of the countries that were misclassified were very close to at least one of the descriptors below.

- [popul(c1) = 17,19..28][sanctions(c1) # 2]³ (10)
- OR [literacy(c1)#2,7..9,15][popul(c1)=1..10] (7)
- OR [polright(c1)=3,5..7][popul(c1) >= 13] (14)
- OR [literacy(c1)=6,10..16,19,20][sanctions(c1)=0] (9)

The rule is specified as a set of disjuncts, any one of which may be satisfied in order for an event to be classified as a low strife country. Each of the disjuncts focuses in on one of the regularities in the sample set. The numbers in parentheses on the right show how many countries in the entire set satisfied that particular disjunct. When developing a theory, these numbers

³Here 19..28 specifies a range of numbers and is equivalent to: 19,20,...,28. The symbol # represents "not equal to".

would become very useful as they would indicate the relative effectiveness of each of the rules. Thus the third disjunct seems to be the most important rule here, covering half of the low strife countries. Paraphrased, it says that a country which has a Political Rights Index of 3, 5, 6, or 7 (at least somewhat suppressive) and whose population density figure is greater than or equal to 13 (average to very densely populated) is to be classified as having low strife.⁴ It is worthwhile to note that the entire rule is much shorter than the descriptor for even one country.⁵

The following is an example of a rule that was generated for countries with high strife. It correctly classified 21 out of the 23 countries with high strife.

[literacy(c1)#2..4,8,14,16,20][litshift(c1)#12..17]	(13)
OR [sanctions(c1)=2]	(4)
OR [popul(c1)#1..4,9,10,16..21,28]	(14)
OR [polstrikes(c1)#0..4]	(6)

The first and third selectors seem to be the most important although it is not clear that anything meaningful can be interpreted from them. The performance of the rule on the test set is somewhat surprising in this context.

The rules that are generated are highly dependent on the particular training set used. If the algorithm discovers a regularity in the training set then a very compact and meaningful rule can be generated. There is a way to increase the chances of this happening. Rather than selecting the sample set randomly, the user might want to choose a specific set of countries according to the value of some relevant variable, such as only those countries in Europe and N. America. One could argue that the probability of finding a simple classification rule for countries which have similar cultures is higher than for randomly picked countries. Moreover such a rule might more helpful in developing novel theories. Astra generated the following rule for European and N. American countries with high strife:

⁴Since strife here is defined as the number of riots and armed attacks per million people, it seems reasonable to conjecture that in certain situations, the more suppressive the government, the less likely the country is to have any incidents.

⁵For an example of how to read a descriptor, see the Appendix.

[polstrikes(c1)#0] (5)
OR [civright(c1)=2..5,7] (7)

Basically this rule says that, if a country had at least one political strike or its civil rights index is between 2 and 5 or is 7, the country is one with high strife. Note that this rule is much simpler than the previous rule for high strife and thus might provide insights into the nature of strife in Western countries.

The second type of rule that Astra generates, characteristic rules, can also provide insights. The following is a characteristic rule for countries in Europe and N. America with high strife:

[civright(c1)#5,6][enshift(c1)=5..7][highered=2..3]
[literacy(c1)=12..20][litshift(c1)=6..11][polright(c1)=1..4]
[popul(c1)=12..20][relaxsanc(c1)=*][sanctions(c1)=*]

The * indicates that the value of a particular variable is irrelevant. Due to the nature of these rules, every European and N. American country with high strife in the database will match the above rule. Although low strife countries may also be covered by the rule, it does indicate what characteristics are *not* present. For example, none of the Western countries with high strife are countries where the political rights are heavily suppressed (polright values of 6 or 7). Several low strife countries do have that characteristic.

7. Discussion

The above results point out some of the strengths and weaknesses of this methodology as compared to some of the more traditional analysis methods, such as correlation studies, regression analysis, and factor analysis. The basic philosophy here is that the process of developing theories is a search process and if the theories can be formulated in logic, this search can be automated to a large extent. Astra provides a formalism for defining the search space and allows the user to specify criteria indicating where to search, and what to look for in a hypothesis. In addition, the completeness and soundness criteria guarantee a minimum performance level for the rules which are found. Due to this framework, there are a number of advantages

that Astra has over the traditional methods of analysis, although it has its share of disadvantages too.

One of the main deficits of the traditional methods is that although some of them can handle multiple variables, they are not able to represent the complex relationships that can sometimes exist between them. Moreover the analysis can become fairly difficult as the number of variables increases. In the formalism presented here these difficulties do not arise. Since Astra's representation language is an extension of predicate calculus, it is possible to represent *any* relationship that can be expressed in predicate logic. Although the computation time does increase as a function of the number of variables⁶, the only conceptual difficulty imposed on the user when a large number of variables is used is actually understanding the rules.

Another disadvantage of the other methods is that, since the representation is strictly numerical, only numerical factors can be considered in developing the theory. In Astra, the values of variables don't have to be numerical. They can represent values which have no particular relationship between them (such as the values of the variable **continent**), or values which have a hierarchical structure to them (such as the classification of political systems.)⁷ Due to this flexibility, many different types of factors may be considered. A further consequence of such a representation language is that the rules which are produced are readable and meaningful to humans. Thus it can be a powerful aid in the formulation of theories as well as in their verification.

Often a simple mathematical relationship can elegantly express the relationship between two quantities. Using the predicate calculus notation does not necessarily prevent the user from discovering these relationships. A feature of Astra that was not used in this study is its ability to derive new variables based on old ones. For example, the user can define a variable **x** to be **sanctions** divided by **relaxsanc**. The system would then add the appropriate selector to any input event whenever it contained a value for

⁶We do not expect this to be a major problem. For this study, Astra typically took around 40 seconds on a Sun 2/150 workstation to generate the rules for both classes.

⁷This ability to represent hierarchical values was not used in this study.

sanctions and **relaxsanc**. Essentially the algorithm would be exploring the feasibility of the equation:

$$\text{sanctions} = x * \text{relaxsanc}$$

The difference here is that **x** is treated just like any other variable and hence can be combined with other selectors to form rules. For example, suppose that all low strife European and North American countries which had a civil rights index of 1 had values for **x** ranging from 1 to 2. Then, the algorithm could induce the following rule to express that relationship:

$$\begin{aligned} [x(c1) = 1,2][\text{continent}(c1) = \text{europe}, \text{namerica}][\text{civright}(c1)=1] \\ \Rightarrow [\text{strife}(c1)= \text{low}] \end{aligned}$$

The disadvantage of finding these relationships in this manner is that the user has to explicitly code in the equations that are to be considered. In the above example, if there was an inverse relationship between **relaxsanc** and **sanctions**, the system would never have discovered it. Dr. Michalski and his colleagues are currently implementing a system called ABACUS which attempts to automatically discover these relationships. Another system, called BACON, built just for inducing mathematical relationships is described in [Langley et. al., 1983]. Among other things this system has been used successfully to derive Kepler's Third Law and Ohm's Law from raw data.

One of the features of Astra is that it attempts to correctly classify *all* of the countries in the sample set. Although this is often a useful property to have, it can lead to less meaningful rules if there are many irregularities in the sample set. More importantly, errors in the measured variables may lead the system to discover more complicated rules than are actually necessary⁸. A possible solution, available in Astra, is to change the requirement that *all* the rules have to be correctly classified to the requirement that *most* of the rules have to be correctly classified, where "most" is a percentage value entered by the user. This method however, still treats each value as if it were exactly correct. Sometimes it might be desirable to associate an error with each value and allow the rules to work within that error.

⁸The rules used in this study attempt to compensate for errors in the data by discretizing the range for each variable into a small number of values.

A problem that arises in this methodology is deciding when a particular rule is a good one. The system attempts to give the user some indication of which rules might be meaningful by printing out how many events are covered by each rule, however, as was indicated in the previous section, this is not always sufficient. It would be interesting and useful if some other measure of the "goodness" of a rule could be calculated. Since the notion of a "good" rule may change with the domain being studied, a flexible, user defined measure might be necessary in order to handle a wide variety of situations. One could then incorporate this measure into the part of the algorithm that influences the selection of possible hypotheses.

A somewhat related problem is that humans, when searching for possible theories, often do not blindly attempt to fit the data to hypotheses. Very often the background knowledge that the user possesses strongly influences the search. For example, a human researcher would know that the lifestyles and attitudes within Western European countries are very similar. Thus, for a human researcher, partitioning the data set into European and non-European countries would be a natural way of searching for regularities in the data. Essentially what is going on is that the researcher is either reducing the number of hypotheses that have to be considered, or changing the order in which they are considered.

There are two benefits gained by doing this. First of all, the amount of time needed to find a hypothesis is drastically reduced. Second, and perhaps more important, the probability of finding a *good* hypothesis is much greater. An improved interface for entering and effectively using domain knowledge would greatly enhance the performance of the system and is an important research area in Machine Learning. Some recent work, taking an extreme position with respect to this issue, is described in [Mitchell, *et al.*, 1986] and [DeJong and Mooney, 1986]. In the methodology described in these papers, it is required that enough domain knowledge be entered into the system such that after presenting *one* positive example, the program is able to construct the correct hypothesis. Although, obviously this approach is too extreme, the papers do make an important point regarding the necessity of effectively utilizing domain knowledge.

8. Conclusion

The results show that techniques developed in Machine Learning can be effectively used as an aid in the analysis of social systems. There are several advantages in using a system such as Astra in developing theories. The language used for representing the hypotheses can be more expressive than numerical methods. The types of variables that can be used are not limited to numerical types. The relationships that can be expressed between the variables are more complex and the hypotheses produced are usually meaningful to humans and reliable. In the case studied, the algorithm produced rules that were excellent predictors, achieving average success rates greater than 80%, while at the same time pointing out some regularities in the sample data that merit further research.

There are a number of disadvantages to the system as it currently stands. The algorithm attempts to correctly classify every sample event; for complex domains this may lead to rules which are not easily interpreted by humans. Errors in the data set can cause similar problems. Finally the means of entering domain knowledge into the system needs to be extended to achieve more flexibility and better results. None of these problems are problems with the formalism itself, but can be solved through further research and by paying attention to the practical aspects of applying these methods.

Perhaps the most important point of the methodology presented in this paper is the philosophy behind it. The procedure is not viewed as one where you simply plug in some variables into a formula, but instead attempts to automate the *search* for a correct theory. In order to this well it is important to have a flexible means of entering domain knowledge to influence the search. The system used in Astra for ordering the possible hypotheses is a start in that direction, however much more needs to be done. Ultimately, we envision a large, flexible system which can be used by scientists as an intelligent tool for the formulation of theories.

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Appendix

The Input Events

This appendix contains all the events in the input format accepted by Astra. The text delimited by '%' are taken to be comments and are ignored by the system. The header comment describes the formulas that were used to derive the values for the different variables. To show how each event is to be read, the descriptor for the first event is paraphrased into English.

% Data for countries for the year 1977.

Explanations of predicates:

strife = (riots + armed attacks)/pop.(millions):

0.000 - 0.299 -> low

> 0.300 -> hi

literacy = Percentage of adults considered literate:

$\text{int}(\text{pct}/5)+1$

popul = Concentration of population:

$10 \cdot \log(\text{conc.})$

sanctions = impositions of sanctions:

> 69 -> 3 (very high)

16 - 69 -> 2 (high)

6 - 15 -> 1 (medium)

0 - 5 -> 0 (low)

highered = Students enrolled in higher education, per million:

> 9000 -> 3 (very high)

4100 - 8999 -> 2 (high)

1500 - 4099 -> 1 (med)

0 - 1499 -> 0 (low)

polright = Political Rights Index:

1 -> Most rights

7 -> Least rights

civright = Civil Rights Index:

1 -> Most rights

7 -> Least rights

polstrikes = Number of political strikes.

relaxsanc = Relaxations of sanctions:

same scale as imposition of sanctions.

continent = continent that a country is in:

one of {namerica,samerica,europe,asia,africa,australia}

prshift = shift in the political rights index: (shift + 6)

crshift = shift in the civil rights index: (shift + 4)

enshift = shift in school enrollment: (pct + 4)

litshift = shift in literacy enrollment: ((pct + 18)/3)

%

%-----Logical Rules-----

mml(e1,e2) -> More Male Labour in e1 than e2

mgdp(e1,e2) -> A larger share of the GDP was due to e1 than e2

The following logical rules specify that the above predicates are transitive.

%

1

[mml(e1,e2)][mml(e2,e3)] => [mml(e1,e3)].

1

[mgdp(e1,e2)][mgdp(e2,e3)] => [mgdp(e1,e3)].

% ----- Descriptors for Countries with Low Strife ----- %
 % Canada %

e
 [literacy(c1)=20][sanctions(c1)=1][popul(c1)=3][highered(c1)=3]
 [mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
 [mgdp(e1,e2)][mgdp(e2,e3)][polright(c1)=1][civright(c1)=1]
 [prshift(c1)=6][crshift(c1)=4][enshift(c1)=5][litshift(c1)=6]
 [continent(c1)=namerica][polstrikes(c1)=0][relaxsanc(c1)=0]
 => [strife=low].

% The above descriptor says that:

IF the literacy value of a particular country is > 95% AND
 the no. of sanctions imposed is between 0 and 5 AND
 the value of popul is 3 AND
 the number of students enrolled in higher education > 9000 AND
 the largest share of the male labour force is in the service sector
 and the smallest is in agriculture AND
 the largest share of the gross domestic product is due to the service
 sector with the smallest share due to agriculture AND
 the political and civil rights indeces for the country are one AND
 there has been no shift in both of these indeces AND
 the shift in school enrollment has been one percent AND
 there has been no shift in literacy AND
 the country is in North America AND
 there have been no political strikes AND
 between 0 and 5 sanctions were relaxed in 1977 THEN
 the country is to be classified as having low strife.

The complexity of the above event descriptor serves to highlight the potential
 of a system which can deal effectively with such rules. %

% Brazil %

e
 [literacy(c1)=14][sanctions(c1)=1][popul(c1)=9][highered(c1)=3]
 [mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
 [polright(c1)=4][civright(c1)=5][mgdp(e1,e2)][mgdp(e2,e3)]
 [prshift(c1)=7][crshift(c1)=5][enshift(c1)=7][litshift(c1)=8]
 [continent(c1)=samerica][polstrikes(c1)=2][relaxsanc(c1)=0]
 => [strife=low].

% Thailand %

e

```
[literacy(c1)= 16][sanctions(c1)= 1 ][popul(c1)= 16][highered(c1)= 1 ]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 5 ][civright(c1)= 4 ][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=11][crshift(c1)=1][enshift(c1)=5][litshift(c1)=9]
[continent(c1)=asia][polstrikes(c1)= 0 ][relaxsanc(c1)= 0 ]
=> [strife= low ].
```

% Belgium %

e

```
[literacy(c1)=20][sanctions(c1)=0][popul(c1)=21][highered(c1)=3]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=1][civright(c1)=1][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=6][crshift(c1)=4][enshift(c1)=4][litshift(c1)=6]
[continent(c1)=europe][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% Uganda %

e

```
[literacy(c1)=7][sanctions(c1)=1][popul(c1)=14][highered(c1)=0]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=7][civright(c1)=7][mgdp(e3,e1)][mgdp(e1,e2)]
[prshift(c1)=6][crshift(c1)=4][enshift(c1)=6][litshift(c1)=9]
[continent(c1)=africa][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% India %

e

```
[literacy(c1)=8][sanctions(c1)=1][popul(c1)=19][highered(c1)=2]
[mml(e3,e2)][mml(e2,e1)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=2][civright(c1)=3][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=5][crshift(c1)=7][enshift(c1)=8][litshift(c1)=9]
[continent(c1)=asia][polstrikes(c1)=1][relaxsanc(c1)=2]
=> [strife=low].
```


% Nepal %

e

```
[literacy(c1)=4][sanctions(c1)=0][popul(c1)=16][higherred(c1)=1]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=6][civright(c1)=5][mgdp(e3,e1)][mgdp(e1,e2)]
[prshift(c1)=6][crshift(c1)=4][enshift(c1)=10][litshift(c1)=10]
[continent(c1)=asia][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% Soviet Union %

e

```
[literacy(c1)=20][sanctions(c1)=2][popul(c1)=9][higherred(c1)=3]
[mml(e2,e1)][mml(e1,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=7][civright(c1)=6][mgdp(e2,e1)][mgdp(e1,e3)]
[prshift(c1)=5][crshift(c1)=4][enshift(c1)=5][litshift(c1)=6]
[continent(c1)=europe,asia][polstrikes(c1)=0][relaxsanc(c1)=1]
=> [strife=low].
```

% Singapore %

e

```
[literacy(c1)=14][sanctions(c1)=0][popul(c1)=28][higherred(c1)=3]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=5][civright(c1)=5][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=6][crshift(c1)=4][enshift(c1)=6][litshift(c1)=9]
[continent(c1)=asia][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% US %

e

```
[literacy(c1)=20][sanctions(c1)=2][popul(c1)=11][higherred(c1)=3]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=1][civright(c1)=1][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=6][crshift(c1)=4][enshift(c1)=4][litshift(c1)=6]
[continent(c1)=namerica][polstrikes(c1)=0][relaxsanc(c1)=1]
=> [strife=low].
```

% Venezuela %

e

```
[literacy(c1)= 17][sanctions(c1)=0 ][popul(c1)= 9][highered(c1)= 3 ]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 2][civright(c1)= 2 ][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=7][crshift(c1)=4][enshift(c1)=6][litshift(c1)=17]
[continent(c1)=samerica][polstrikes(c1)= 0 ][relaxsanc(c1)= 0 ]
=> [strife= low].
```

% North Korea %

e

```
[sanctions(c1)= 0 ][popul(c1)= 18]
[mml(e3,e2)][mml(e2,e1)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 7 ][civright(c1)= 7 ]
[prshift(c1)=6][crshift(c1)=4]
[continent(c1)=asia][polstrikes(c1)= 0 ][relaxsanc(c1)= 0 ]
=> [strife= low].
```

% Sweden %

e

```
[literacy(c1)= 20][sanctions(c1)= 0 ][popul(c1)= 10][highered(c1)= 3 ]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 1][civright(c1)=1 ][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=5][crshift(c1)=4][enshift(c1)=4][litshift(c1)=6]
[continent(c1)=europe][polstrikes(c1)= 0 ][relaxsanc(c1)= 0 ]
=> [strife= low].
```

% Japan %

e

```
[literacy(c1)=20][sanctions(c1)=0][popul(c1)=20][highered(c1)=3]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=2][civright(c1)=1][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=6][crshift(c1)=4][enshift(c1)=4][litshift(c1)=6]
[continent(c1)=asia][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% Hungary %

```
e
[literacy(c1)=20][sanctions(c1)=0][popul(c1)=17][highered(c1)=3]
[mml(e2,e1)][mml(e1,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=6][civright(c1)=6][mgdp(e2,e1)][mgdp(e1,e3)]
[prshift(c1)=6][crshift(c1)=5][enshift(c1)=5][litshift(c1)=6]
[continent(c1)=europe][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% Austria %

```
e
[literacy(c1)=20][sanctions(c1)=0][popul(c1)=16][highered(c1)=3]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=1][civright(c1)=1][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=6][crshift(c1)=4][enshift(c1)=5][litshift(c1)=6]
[continent(c1)=europe][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% S. Korea %

```
e
[literacy(c1)=18][sanctions(c1)=0][popul(c1)=21][highered(c1)=2]
[mml(e3,e2)][mml(e2,e1)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=5][civright(c1)=6][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=7][crshift(c1)=5][enshift(c1)=6][litshift(c1)=12]
[continent(c1)=asia][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% Chad %

```
e
[literacy(c1)= 3 ][sanctions(c1)= 0 ][popul(c1)= 4 ][highered(c1)= 0 ]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 6 ][civright(c1)= 6 ][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=5][crshift(c1)=5][enshift(c1)=15][litshift(c1)=9]
[continent(c1)=africa][polstrikes(c1)= 4 ][relaxsanc(c1)= 0 ]
=> [strife= low ].
```

% Nigeria %

```
e
[literacy(c1)=4][sanctions(c1)=0][popul(c1)=15][higherred(c1)=0]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=6][civright(c1)=4][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=7][crshift(c1)=6][enshift(c1)=7][litshift(c1)=0]
[continent(c1)=africa][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% West Germany %

```
e
[literacy(c1)=20][sanctions(c1)=1][popul(c1)=20][higherred(c1)=3]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=1][civright(c1)=1][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=6][crshift(c1)=3][enshift(c1)=4][litshift(c1)=6]
[continent(c1)=europe][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% China %

```
e
[continent(c1)=asia][literacy(c1)=17][sanctions(c1)=0][popul(c1)= 16]
[mml(e3,e2)][mml(e2,e1)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=7][civright(c1)=7]
[prshift(c1)=7][crshift(c1)=5][enshift(c1)=12][litshift(c1)=17]
[polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% Poland %

```
e
[literacy(c1)=20][sanctions(c1)=1][popul(c1)=17][higherred(c1)=3]
[mml(e2,e3)][mml(e3,e1)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=6][civright(c1)=6][mgdp(e2,e1)][mgdp(e1,e3)]
[prshift(c1)=6][crshift(c1)=5][enshift(c1)=5][litshift(c1)=6]
[continent(c1)=europe][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% Switzerland %

e

```
[literacy(c1)=20][sanctions(c1)=0][popul(c1)=18][highered(c1)=3]
[mml(e2,e1)][mml(e1,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=1][civright(c1)=1]
[prshift(c1)=6][crshift(c1)=4][enshift(c1)=4][litshift(c1)=6]
[continent(c1)=europe][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% Morocco %

e

```
[literacy(c1)=5][sanctions(c1)=0][popul(c1)=13][highered(c1)=1]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=5][civright(c1)=4][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=8][crshift(c1)=6][enshift(c1)=8][litshift(c1)=9]
[continent(c1)=africa][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% Haiti %

e

```
[literacy(c1)= 5 ][sanctions(c1)= 0 ][popul(c1)= 18][highered(c1)= 0 ]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 6 ][civright(c1)= 6 ][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=7][crshift(c1)=4][enshift(c1)=6][litshift(c1)=10]
[continent(c1)= namerica,samerica][polstrikes(c1)= 0 ][relaxsanc(c1)= 0 ]
=> [strife= low].
```

% Kenya %

e

```
[literacy(c1)=8][sanctions(c1)=0][popul(c1)=11][highered(c1)=0]
[mml(e3,e2)][mml(e2,e1)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=5][civright(c1)=5][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=6][crshift(c1)=3][enshift(c1)=9][litshift(c1)=12]
[continent(c1)=africa][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% Malaysia %

e

```
[literacy(c1)=12][sanctions(c1)=0][popul(c1)=13][highered(c1)=1]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=3][civright(c1)=3][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=5][crshift(c1)=3][enshift(c1)=6][litshift(c1)=11]
[continent(c1)=asia][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% Indonesia %

e

```
[literacy(c1)=13][sanctions(c1)=0][popul(c1)=15][highered(c1)=1]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=5][civright(c1)=5][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=6][crshift(c1)=4][enshift(c1)=7][litshift(c1)=12]
[continent(c1)=asia][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=low].
```

% ----- Descriptors for Countries with High Strife ----- %

% Zimbabwe %

e

```
[literacy(c1)= 8 ][sanctions(c1)= 0 ][popul(c1)= 10]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 6 ][civright(c1)= 5 ][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=7][crshift(c1)=4][enshift(c1)=5][litshift(c1)=12]
[continent(c1)=africa][polstrikes(c1)= 0 ][relaxsanc(c1)= 0 ]
=> [strife= hi].
```

% Pakistan %

e

```
[literacy(c1)=5][sanctions(c1)=2][popul(c1)=16][highered(c1)=1]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=4][civright(c1)=5][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=3][crshift(c1)=5]
[continent(c1)=asia][polstrikes(c1)=5][relaxsanc(c1)=2]
=> [strife=hi].
```

% Greece %

```
e
[literacy(c1)=17][sanctions(c1)=1][popul(c1)=15][highered(c1)=3]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=3][civright(c1)=3][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=11][crshift(c1)=8][enshift(c1)=5][litshift(c1)=7]
[continent(c1)=europe][polstrikes(c1)=1][relaxsanc(c1)=0]
=> [strife=hi].
```

% Angola %

```
e
[sanctions(c1)= 0 ][popul(c1)= 6 ]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[enshift(c1)=21]
[continent(c1)=africa][polstrikes(c1)= 0 ][relaxsanc(c1)= 0 ]
=> [strife= hi].
```

% Portugal %

```
e
[literacy(c1)=15][sanctions(c1)=0][popul(c1)=16][highered(c1)=3]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=4][civright(c1)=3][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=9][crshift(c1)=8][enshift(c1)=7][litshift(c1)=9]
[continent(c1)=europe][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=hi].
```

% Zambia %

```
e
[literacy(c1)=8][sanctions(c1)=0][popul(c1)=7][highered(c1)=1]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=5][civright(c1)=5][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=6][crshift(c1)=5][enshift(c1)=8][litshift(c1)=5]
[continent(c1)=africa][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=hi].
```

% Mexico %

e

```
[literacy(c1)=15][sanctions(c1)=1][popul(c1)=12][highered(c1)=2]
[mml(e1,e3)][mml(e3,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=4][civright(c1)=3][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=7][crshift(c1)=3][enshift(c1)=7][litshift(c1)=9]
[continent(c1)=namerica][polstrikes(c1)=3][relaxsanc(c1)=0]
=> [strife=hi].
```

% United Kingdom %

e

```
[literacy(c1)=20][sanctions(c1)=2][popul(c1)=20][highered(c1)=3]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=1][civright(c1)=1][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=6][crshift(c1)=4][enshift(c1)=5][litshift(c1)=6]
[continent(c1)=europe][polstrikes(c1)=10][relaxsanc(c1)=0]
=> [strife=hi].
```

% Argentina %

e

```
[literacy(c1)=19][sanctions(c1)=1][popul(c1)=8][highered(c1)=3]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=4][civright(c1)=4][mgdp(e1,e2)][mgdp(e2,e3)]
[continent(c1)=europe][polstrikes(c1)=0][relaxsanc(c1)=4]
[prshift(c1)=10][crshift(c1)=0][enshift(c1)=5][litshift(c1)=17]
=> [strife=hi].
```

% France %

e

```
[literacy(c1)=20][sanctions(c1)=1][popul(c1)=16][highered(c1)=3]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=1][civright(c1)=2][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=6][crshift(c1)=5][enshift(c1)=5][litshift(c1)=6]
[continent(c1)=europe][polstrikes(c1)=6][relaxsanc(c1)=0]
=> [strife=hi].
```


% Libya %

e

```
[literacy(c1)=9][sanctions(c1)=0][popul(c1)=1][highered(c1)=2]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=7][civright(c1)=6][mgdp(e2,e1)][mgdp(e1,e3)]
[prshift(c1)=7][crshift(c1)=3][enshift(c1)=13][litshift(c1)=11]
[continent(c1)=africa][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=hi].
```

% Lebanon %

e

```
[sanctions(c1)= 1 ][popul(c1)= 20]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 3 ][civright(c1)= 3 ][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=4][crshift(c1)=2][enshift(c1)=5]
[continent(c1)=asia][polstrikes(c1)= 6 ][relaxsanc(c1)= 0 ]
=> [strife= hi].
```

% Ethiopia %

e

```
[literacy(c1)= 2 ][sanctions(c1)= 1 ][popul(c1)= 11][highered(c1)= 0 ]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 6 ][civright(c1)= 6 ][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=4][crshift(c1)=2][enshift(c1)=14][litshift(c1)=8]
[continent(c1)=africa][polstrikes(c1)= 0 ][relaxsanc(c1)= 0 ]
=> [strife= hi].
```

% Turkey %

e

```
[literacy(c1)= 12][sanctions(c1)= 1 ][popul(c1)= 14][highered(c1)= 2]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 2 ][civright(c1)= 3 ][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=7][crshift(c1)=5][enshift(c1)=7][litshift(c1)=11]
[continent(c1)=asia,europe][polstrikes(c1)= 0 ][relaxsanc(c1)= 0 ]
=> [strife= hi].
```

% Spain %

e

```
[literacy(c1)=18][sanctions(c1)=3][popul(c1)=15][highered(c1)=3]
[mml(e2,e1)][mml(e1,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=4][civright(c1)=4][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=9][crshift(c1)=8][enshift(c1)=6][litshift(c1)=7]
[continent(c1)=europe][polstrikes(c1)=10][relaxsanc(c1)=3]
=> [strife=hi].
```

% South Africa %

e

```
[literacy(c1)=12][sanctions(c1)=2][popul(c1)= 11][highered(c1)=1]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=4][civright(c1)=5][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=5][crshift(c1)=3][enshift(c1)=6][litshift(c1)=13]
[continent(c1)=africa][polstrikes(c1)=3][relaxsanc(c1)=0]
=> [strife=hi].
```

% Peru %

e

```
[literacy(c1)= 15 ][sanctions(c1)= 1 ][popul(c1)= 9 ][highered(c1)= 3 ]
[mml(e1,e3)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 6 ][civright(c1)= 5 ][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=8][crshift(c1)=6][enshift(c1)=7][litshift(c1)=10]
[continent(c1)=samerica][polstrikes(c1)= 7][relaxsanc(c1)= 0]
=> [strife= hi].
```

% Iraq %

e

```
[literacy(c1)= 5 ][sanctions(c1)= 0 ][popul(c1)= 12][highered(c1)= 2 ]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 7 ][civright(c1)= 7 ][mgdp(e2,e1)][mgdp(e1,e3)]
[prshift(c1)=6][crshift(c1)=5][enshift(c1)=9][litshift(c1)=7]
[continent(c1)=asia][polstrikes(c1)= 0 ][relaxsanc(c1)= 0 ]
=> [strife= hi].
```

% Colombia %

e

```
[literacy(c1)=17][sanctions(c1)=0][popul(c1)=11][highered(c1)=2]
[mml(e1,e3)][mml(e3,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=2][civright(c1)=3][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=6][crshift(c1)=3][enshift(c1)=7][litshift(c1)=11]
[continent(c1)=samerica][polstrikes(c1)=1][relaxsanc(c1)=0]
=> [strife=hi].
```

% Zaire %

e

```
[literacy(c1)= 7 ][sanctions(c1)= 0 ][popul(c1)= 8 ][highered(c1)= 0 ]
[mml(e3,e2)][mml(e2,e1)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 7 ][civright(c1)= 6 ][mgdp(e1,e3)][mgdp(e3,e2)]
[prshift(c1)=6][crshift(c1)=2][enshift(c1)=7][litshift(c1)=5]
[continent(c1)=africa][polstrikes(c1)= 0 ][relaxsanc(c1)= 0 ]
=> [strife= hi].
```

% Burma %

e

```
[literacy(c1)=14][sanctions(c1)=4][popul(c1)=14][highered(c1)=1]
[mml(e3,e1)][mml(e1,e2)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=7][civright(c1)=6][mgdp(e3,e1)][mgdp(e1,e2)]
[prshift(c1)=7][crshift(c1)=3][enshift(c1)=10][litshift(c1)=8]
[continent(c1)=asia][polstrikes(c1)=0][relaxsanc(c1)=0]
=> [strife=hi].
```

% Italy %

e

```
[literacy(c1)=20][sanctions(c1)=2][popul(c1)=19][highered(c1)=3]
[mml(e2,e1)][mml(e1,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)=1][civright(c1)=2][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=5][crshift(c1)=4][enshift(c1)=6][litshift(c1)=8]
[continent(c1)=europe][polstrikes(c1)=0][relaxsanc(c1)=5]
=> [strife=hi].
```

% Israel %

e

```
[literacy(c1)= 18][sanctions(c1)= 0 ][popul(c1)= 18][highered(c1)= 3 ]
[mml(e1,e2)][mml(e2,e3)][kind(e1)=services][kind(e2)=indus][kind(e3)=agr]
[polright(c1)= 2 ][civright(c1)= 3 ][mgdp(e1,e2)][mgdp(e2,e3)]
[prshift(c1)=6][crshift(c1)=5][enshift(c1)=4][litshift(c1)=5]
[continent(c1)=asia][polstrikes(c1)= 4 ][relaxsanc(c1)= 0 ]
=> [strife= hi].
```